

C Programming Lab – Test Cases Manual

1. Fibonacci Number Using Recursion

Test Case ID	Input	Expected Output
TC1	n = 0	Fibonacci = 0
TC2	n = 1	Fibonacci = 1
TC3	n = 5	Fibonacci = 5
TC4	n = 10	Fibonacci = 55
TC5	n = -3	Invalid input message

2. GCD of Three Numbers Using Recursion

Test Case ID	Input	Expected Output
TC1	12, 18, 24	GCD = 6
TC2	7, 14, 21	GCD = 7
TC3	5, 10, 15	GCD = 5
TC4	9, 27, 81	GCD = 9
TC5	0, 10, 20	GCD = 10

3. Pascal's Triangle

Test Case ID	Input	Expected Output
TC1	rows = 1	1
TC2	rows = 3	1 / 1 1 / 1 2 1
TC3	rows = 5	Correct Pascal triangle with 5 rows
TC4	rows = 0	Invalid input
TC5	rows = 8	Triangle generated correctly

4. Dynamic Memory Allocation

Test Case ID	Operation	Expected Output
TC1	malloc() for 5 integers	Memory allocated successfully
TC2	calloc() for 5 integers	Elements initialized to 0
TC3	realloc() increase size	New size updated successfully
TC4	free()	Memory released
TC5	Allocation failure	Error message displayed

5. Call by Value and Call by Reference

Test Case ID	Input	Expected Output
TC1	a=10, b=20	Values unchanged in call by value
TC2	a=10, b=20	Values swapped in call by reference
TC3	a=0, b=0	Output = 0,0
TC4	Negative values	Correct swap
TC5	Large integers	Correct operation

6. Type Casting in C

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

7. String Operations Using Functions

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

8. String Operations Without Built-in Functions

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

9. Delete Element in Array

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

10. Insert Element in Array

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

11. Remove Duplicates from Sorted Array

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

12. Kth Smallest Element

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

13. Modify Array Elements Using Pointers

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

14. String Length Using Pointers

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

15. Generate Pseudo-Random Numbers

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

16. Prime Factors of Integer

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

17. Merge Two Files

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

18. Character, Word and Line Count

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

19. Longthiest Line in Text File

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

20. Reverse First n Characters in File

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.

General Validation Test Cases

Includes multiple functional, boundary, invalid input, and edge case test scenarios for validation.